Vector Spaces V: Bases, Dimension and Polynomial Spaces

From the class book: Chapter 3, Section 3.4 (with coordinates from 4.4)

1. Key definitions

Basis. A basis of V is a linearly independent spanning set. Equivalently, it is a maximal independent set, a minimal spanning set, or a set β such that every $v \in V$ is a *unique* combination of elements of β.

Dimension and coordinates. All bases of V have the same size, $\dim V$. For an ordered basis $\beta = \{v_1, \ldots, v_n\}$ and $v = a_1 v_1 + \cdots + a_n v_n$, the coordinate vector is $[v]_\beta = (a_1, \ldots, a_n)^t$, and $v \mapsto [v]_\beta$ is an isomorphism $V \to F^n$.

- P_n is the space of real polynomials of degree $\leq n$; its standard basis is $\{1, x, \ldots, x^n\}$, so $\dim P_n = n + 1$.
- To test whether polynomials form a basis, write their coefficient vectors as rows of a matrix and compute its rank.

2. Main results

Theorem (Dimension theorems, $\dim V = n$)

- An independent set has at most n vectors and a spanning set at least n. Exactly n independent vectors, or exactly n spanning vectors, form a basis.
- Every independent set extends to a basis, and every spanning set contains one.
- A subspace W has $\dim W \leq n$, with equality iff $W = V$.
- A subspace of F^n cut out by k independent homogeneous linear equations has dimension $n - k$.

Space	Dimension
P_n	$n+1$
polynomials of degree $\le n$ in 2 variables; in 3 variables	$\binom{n+2}{2}$; $\binom{n+3}{3}$
homogeneous polynomials of degree d in k variables	$\binom{d+k-1}{k-1}$
$M_{m\times n}(F)$	mn
symmetric; skew-symmetric $n \times n$	$\frac{n(n+1)}{2}$; $\frac{n(n-1)}{2}$
upper triangular; diagonal	$\frac{n(n+1)}{2}$; n
trace zero $n \times n$	n^2-1
Hermitian (over $\mathbb{R}$)	n^2
all row and column sums 0	$(n-1)^2$

Change of basis

- If the columns of P are the vectors of β' written in the basis β, then $[v]_\beta = P[v]_{\beta'}$, and $[T]_{\beta'} = P^{-1}[T]_\beta P$.
- On P_n, Taylor's formula gives the coordinates in the basis $\{1, (x-a), \dots, (x-a)^n\}$: $p = \sum_k \frac{p^{(k)}(a)}{k!}(x-a)^k$.

3. Methods

Method: dimension of a subspace of polynomials.

- (1) Write $p = c_0 + c_1 x + \cdots + c_n x^n$ and turn every condition ($p(a)=0$, $p'(a)=0$, p even, $\int p = 0$) into a linear equation in the c_i.
- (2) The dimension is $n+1$ minus the rank of these equations.
- (3) Conditions $p(a_1) = \cdots = p(a_k) = 0$ at distinct points are independent when $k \le n+1$ (Vandermonde), and then $W = (x-a_1)\cdots(x-a_k)P_{n-k}$.

Method: is a given set a basis?

- Check that the number of vectors equals $\dim V$, then check independence by a determinant or rank.
- Look for an obvious relation such as alternating sums: $(1+x)-(x+x^2)+(x^2+x^3)-(x^3+1)=0$.
- Polynomials of distinct degrees $0, 1, \dots, n$ always form a basis of P_n.

4. Exam shortcuts and common traps

- Even polynomials in P_{2m} have dimension $m+1$, and odd ones m.

- $\{p \in P_n : p(a) = 0\}$ has dimension n, with basis $\{(x-a), (x-a)^2, \ldots, (x-a)^n\}$ or $\{(x-a), x(x-a), \ldots, x^{n-1}(x-a)\}$.
- $\operatorname{span}\{\sin^2 x, \cos^2 x, \cos 2x, 1\}$ has dimension 2, since $\cos 2x = \cos^2 x - \sin^2 x$ and $1 = \cos^2 x + \sin^2 x$.
- **Traps:** $\dim P_n = n+1$, not n; dimension over $\mathbb{R}$ of a complex space doubles; for symmetric matrices the row-sum and column-sum conditions coincide, so there are only n of them.

5. Solved examples

Example 1. Find the coordinates of x^2 in the basis $\{1, x-1, (x-1)^2\}$ of P_2.

Solution

- By Taylor at $a = 1$: $x^2 = 1 + 2(x-1) + (x-1)^2$.
- So $[x^2] = (1, 2, 1)$.

Example 2. Find $\dim W$ for $W = \{p \in P_6 : p(-x) = p(x),\ p(1) = 0\}$.

Solution

- The even polynomials in P_6 are $\operatorname{span}\{1, x^2, x^4, x^6\}$, of dimension 4.
- $p(1) = 0$ is one more independent condition, so $\dim W = 3$.

Example 3. Find the dimension of the space of homogeneous polynomials of degree 4 in x, y, z.

Solution

- The monomials $x^a y^b z^c$ with $a + b + c = 4$ form a basis.
- Their number is $\binom{4+2}{2} = 15$.

Example 4. Find the dimension of the space of 3×3 real symmetric matrices with every row sum 0.

Solution

- Symmetric matrices: 6 parameters.
- The three row-sum conditions are independent (and the column sums then vanish automatically), so the dimension is $6 - 3 = 3$.

Example 5. Decide which of the following are bases of P_3: $\{1+x,\ x+x^2,\ x^2+x^3,\ x^3+1\}$ and $\{x^3,\ x^3+x^2,\ x^3+x^2+x,\ x^3+x^2+x+1\}$.

Solution

- The first set satisfies $v_1 - v_2 + v_3 - v_4 = 0$, so it is not a basis (its rank is 3).
- The second is triangular in the standard basis with 1s on the diagonal, so it is a basis.

Example 6. Let $W = \{p \in P_4 : p(1) = 0\}$. Give two bases of W and find $\dim(W \cap \{\text{even}\})$.

Solution

- $\dim W = 4$, with bases $\{(x-1)^k : 1 \le k \le 4\}$ and $\{x^k(x-1) : 0 \le k \le 3\}$.
- Even polynomials $a + bx^2 + cx^4$ with $a + b + c = 0$: dimension 2.